Chapter 5. Manifolds

Volumes of parallelopipeds and parametrized-manifolds, manifolds in $\mathbb{R}^n$ and their boundaries, integrating scalar functions

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Overview

For a bounded rectifiable set $A \subset \mathbb{R}^k$, $\mathrm{v}(A) = \int_A 1$ (length for $k = 1$, area for $k = 2$). Calculus also measures the length of curves and the area of surfaces in $\mathbb{R}^3$. This chapter introduces their k-dimensional analogues, *k-manifolds in $\mathbb{R}^n$*, defines k-dimensional volume for them, and defines the integral of a scalar function over a k-manifold with respect to volume.

§21. The Volume of a Parallelopiped

Let $\mathcal{P}$ be a k-dimensional parallelopiped in $\mathbb{R}^n$, $k < n$. Its n-dimensional volume is zero, since it lies in a k-dimensional subspace, which has measure zero in $\mathbb{R}^n$. A reasonable k-dimensional volume should (1) be preserved by orthogonal transformations of $\mathbb{R}^n$, and (2) agree with the usual volume when $\mathcal{P}$ lies in $\mathbb{R}^k \times 0$. These conditions determine it.

Lemma 21.1

Let W be a linear subspace of $\mathbb{R}^n$ of dimension k. Then there is an orthonormal basis for $\mathbb{R}^n$ whose first k elements form a basis for W.

Proof. By Theorem 1.2 there is a basis $\mathbf{a}_1, \dots, \mathbf{a}_n$ of $\mathbb{R}^n$ whose first k elements form a basis of W. The *Gram–Schmidt process* produces an orthogonal set $\mathbf{b}_1, \dots, \mathbf{b}_n$ such that for each i, $\mathbf{b}_1, \dots, \mathbf{b}_i$ span the same space as $\mathbf{a}_1, \dots, \mathbf{a}_i$: set $\mathbf{b}_1 = \mathbf{a}_1$ and in general

$$\mathbf{b}_i = \mathbf{a}_i - \lambda_{i1}\mathbf{b}_1 - \lambda_{i2}\mathbf{b}_2 - \cdots - \lambda_{i,i-1}\mathbf{b}_{i-1}.$$

Whatever the scalars λ_{ij}, each $\mathbf{a}_j$ is a combination of $\mathbf{b}_1, \dots, \mathbf{b}_j$ and each $\mathbf{b}_j$ is a combination of $\mathbf{a}_1, \dots, \mathbf{a}_j$ (induction), so $\mathbf{a}_1, \dots, \mathbf{a}_i$ and $\mathbf{b}_1, \dots, \mathbf{b}_i$ span the same subspace; in particular the n vectors $\mathbf{b}_i$ span $\mathbb{R}^n$, so they are independent and nonzero. If $\mathbf{b}_1, \dots, \mathbf{b}_{i-1}$ are mutually orthogonal, taking the dot product of the equation for $\mathbf{b}_i$ with $\mathbf{b}_j$ $(j < i)$ gives $\langle \mathbf{b}_i, \mathbf{b}_j \rangle = \langle \mathbf{a}_i, \mathbf{b}_j \rangle - \lambda_{ij}\langle \mathbf{b}_j, \mathbf{b}_j \rangle$; since $\mathbf{b}_j \neq \mathbf{0}$, a unique λ_{ij} makes this vanish. With these choices the $\mathbf{b}_i$ are mutually orthogonal. Dividing each by its norm gives the desired orthonormal basis. ■

Theorem 21.2

Let W be a k-dimensional linear subspace of $\mathbb{R}^n$. There is an orthogonal transformation $h : \mathbb{R}^n \to \mathbb{R}^n$ that carries W onto the subspace $\mathbb{R}^k \times 0$ of $\mathbb{R}^n$.

Proof. Choose an orthonormal basis $\mathbf{b}_1, \dots, \mathbf{b}_n$ of $\mathbb{R}^n$ with $\mathbf{b}_1, \dots, \mathbf{b}_k$ a basis of W. Let $g(\mathbf{x}) = B \cdot \mathbf{x}$, where B has columns $\mathbf{b}_1, \dots, \mathbf{b}_n$; g is orthogonal and $g(\mathbf{e}_i) = \mathbf{b}_i$, so g carries $\mathbb{R}^k \times 0$ (basis $\mathbf{e}_1, \dots, \mathbf{e}_k$) onto W. Its inverse is the desired map. ■

Theorem 21.3

There is a unique function V that assigns, to each k-tuple $(\mathbf{x}_1, \dots, \mathbf{x}_k)$ of elements of $\mathbb{R}^n$, a non-negative number such that:

(1) If $h : \mathbb{R}^n \to \mathbb{R}^n$ is an orthogonal transformation, then $V(h(\mathbf{x}_1), \dots, h(\mathbf{x}_k)) = V(\mathbf{x}_1, \dots, \mathbf{x}_k)$.

(2) If $\mathbf{y}_1, \dots, \mathbf{y}_k$ belong to the subspace $\mathbb{R}^k \times 0$ of $\mathbb{R}^n$, so that $\mathbf{y}_i = \begin{bmatrix} \mathbf{z}_i \\ 0 \end{bmatrix}$ for $\mathbf{z}_i \in \mathbb{R}^k$, then $V(\mathbf{y}_1, \dots, \mathbf{y}_k) = |\det[\mathbf{z}_1 \ \cdots \ \mathbf{z}_k]|$.

The function V vanishes if and only if the vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$ are dependent. It satisfies the equation

$$V(\mathbf{x}_1, \dots, \mathbf{x}_k) = [\det(X^{\mathrm{tr}} \cdot X)]^{1/2},$$

where X is the n by k matrix $X = [\mathbf{x}_1 \ \cdots \ \mathbf{x}_k]$.

We often write $V(\mathbf{x}_1, \dots, \mathbf{x}_k)$ simply as $V(X)$.

Proof. Let $F(X) = \det(X^{\mathrm{tr}} \cdot X)$.

Step 1. If $h(\mathbf{x}) = A \cdot \mathbf{x}$ with A orthogonal, $F(A \cdot X) = \det(X^{\mathrm{tr}} A^{\mathrm{tr}} A X) = \det(X^{\mathrm{tr}} X) = F(X)$. If Z is k by k and $Y = \begin{bmatrix} Z \\ 0 \end{bmatrix}$, then $F(Y) = \det([Z^{\mathrm{tr}} \ 0] \cdot \begin{bmatrix} Z \\ 0 \end{bmatrix}) = \det(Z^{\mathrm{tr}} Z) = \det^2 Z$.

Step 2. F is non-negative: given $\mathbf{x}_1, \dots, \mathbf{x}_k$, let W be a k-dimensional subspace containing them, and $h(\mathbf{x}) = A \cdot \mathbf{x}$ an orthogonal transformation carrying W onto $\mathbb{R}^k \times 0$ (Theorem 21.2). Then $A \cdot X = \begin{bmatrix} Z \\ 0 \end{bmatrix}$ and $F(X) = F(A \cdot X) = \det^2 Z \geq 0$. Also $F(X) = 0$ iff the columns of Z are dependent, iff $\mathbf{x}_1, \dots, \mathbf{x}_k$ are dependent.

Step 3. Define $V(X) = (F(X))^{1/2}$. By Step 1, V satisfies (1) and (2), and by Step 2's computation V is uniquely determined by them. ■

Definition 21.1 (k-volume of a parallelopiped)

If $\mathbf{x}_1, \dots, \mathbf{x}_k$ are independent vectors in $\mathbb{R}^n$, the **k-dimensional volume** of $\mathcal{P} = \mathcal{P}(\mathbf{x}_1, \dots, \mathbf{x}_k)$ is the positive number $V(\mathbf{x}_1, \dots, \mathbf{x}_k)$.

Example 21.1

Let $\mathbf{a}, \mathbf{b}$ be independent in $\mathbb{R}^3$, $X = [\mathbf{a} \ \mathbf{b}]$, and θ the angle between them, $\langle \mathbf{a}, \mathbf{b} \rangle = \|\mathbf{a}\| \|\mathbf{b}\| \cos\theta$. Then

$$(V(X))^2 = \det \begin{bmatrix} \|\mathbf{a}\|^2 & \langle \mathbf{a}, \mathbf{b} \rangle \\ \langle \mathbf{b}, \mathbf{a} \rangle & \|\mathbf{b}\|^2 \end{bmatrix} = \|\mathbf{a}\|^2 \|\mathbf{b}\|^2 (1 - \cos^2\theta) = \|\mathbf{a}\|^2 \|\mathbf{b}\|^2 \sin^2\theta,$$

the square of the area (base $\|\mathbf{a}\|$ times height $\|\mathbf{b}\| \sin\theta$) of the parallelogram with edges $\mathbf{a}, \mathbf{b}$.

Calculus also gives the area of $\mathcal{P}(\mathbf{a}, \mathbf{b})$ as $\|\mathbf{a} \times \mathbf{b}\|$, where

$$\mathbf{a} \times \mathbf{b} = \det \begin{bmatrix} a_2 & b_2 \\ a_3 & b_3 \end{bmatrix} \mathbf{e}_1 - \det \begin{bmatrix} a_1 & b_1 \\ a_3 & b_3 \end{bmatrix} \mathbf{e}_2 + \det \begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix} \mathbf{e}_3,$$

justified by verifying $\|\mathbf{a}\|^2 \|\mathbf{b}\|^2 - \langle \mathbf{a}, \mathbf{b} \rangle^2 = \|\mathbf{a} \times \mathbf{b}\|^2$. Theorem 21.3 is convenient in theory; the following generalization of the cross-product formula is often more convenient in practice.

Definition 21.2 (ascending k-tuple)

Let $X = [\mathbf{x}_1 \ \cdots \ \mathbf{x}_k]$ be an n by k matrix, $k \leq n$. If $I = (i_1, \dots, i_k)$ with $1 \leq i_1 < i_2 < \cdots < i_k \leq n$, I is an **ascending k-tuple** from $\{1, \dots, n\}$, and X_I or $X(i_1, \dots, i_k)$ denotes the k by k submatrix of X consisting of rows $i_1, \dots, i_k$. More generally, for any k-tuple I from $\{1, \dots, n\}$ (not necessarily distinct or ordered), X_I denotes the k by k matrix whose successive rows are rows $i_1, \dots, i_k$ of X.

Theorem 21.4

Let X be an n by k matrix with $k \le n$. Then

$$V(X) = \Big[\sum_{[I]} \det{}^2 X_I\Big]^{1/2},$$

where the symbol $[I]$ indicates that the summation extends over all ascending k-tuples from the set $\{1, \dots, n\}$.

This is a "Pythagorean theorem for k-volume": the square of the volume of $\mathcal{P}$ is the sum of the squares of the volumes of its projections onto the coordinate k-planes.

Proof. Let $F(X) = \det(X^{\mathrm{tr}}X)$ and $G(X) = \sum_{[I]} \det^2 X_I$; we show $F = G$.

Step 1. If $k = 1$, X is a column with entries $\lambda_1, \dots, \lambda_n$ and $F(X) = \sum \lambda_i^2 = G(X)$. If $k = n$, G has one term and $F(X) = \det^2 X = G(X)$.

Step 2. If the columns $\mathbf{x}_i$ are orthogonal, $X^{\mathrm{tr}}X$ is diagonal with entries $\|\mathbf{x}_i\|^2$, so $F(X) = \|\mathbf{x}_1\|^2 \cdots \|\mathbf{x}_k\|^2$.

Step 3. Neither (1) exchanging columns $j \neq l$ nor (2) replacing column j by itself plus c times column l changes F or G. Such a column operation transforms X into $X \cdot E^{\mathrm{tr}}$, where E is the corresponding elementary matrix, with $\det E = \pm 1$. So $F(XE^{\mathrm{tr}}) = \det(EX^{\mathrm{tr}}XE^{\mathrm{tr}}) = F(X)$. Deleting rows commutes with column operations, so $(XE^{\mathrm{tr}})_I = X_I E^{\mathrm{tr}}$, and $G(XE^{\mathrm{tr}}) = \sum_{[I]} \det^2(X_I) \det^2(E^{\mathrm{tr}}) = G(X)$.

Step 4. It suffices to prove the theorem when all entries of the bottom row of X are zero except possibly the last, and the columns are orthogonal. Given X, if the last row has a nonzero entry, column operations of the above types bring X to the form whose last row is $(0, \dots, 0, \lambda)$, $\lambda \neq 0$; if the last row is zero this holds with $\lambda = 0$. Then apply Gram–Schmidt to the columns: the jth column is replaced by itself minus multiples of earlier columns—operations of type (2)—which keep the zeros in the last row. Afterwards the columns are orthogonal and the matrix still has this form.

Step 5. Induction on n. For $n = 1$, $k = 1$ (Step 1); for $n = 2$, $k = 1$ or 2 (Step 1). Assume the result for fewer than n rows and let $1 < k < n$. By Step 4 we may assume

$$X = \begin{bmatrix} \mathbf{b}_1 & \cdots & \mathbf{b}_{k-1} & \mathbf{b}_k \\ 0 & \cdots & 0 & \lambda \end{bmatrix}$$

with orthogonal columns; the $\mathbf{b}_i \in \mathbb{R}^{n-1}$ are then orthogonal. Let $B = [\mathbf{b}_1 \cdots \mathbf{b}_k]$, $C = [\mathbf{b}_1 \cdots \mathbf{b}_{k-1}]$. By Step 2,

$$F(X) = \|\mathbf{b}_1\|^2 \cdots \|\mathbf{b}_{k-1}\|^2(\|\mathbf{b}_k\|^2 + \lambda^2) = F(B) + \lambda^2 F(C).$$

Split $G(X) = \sum_{i_k<n} \det^2 X_I + \sum_{i_k=n} \det^2 X_I$. If $i_k < n$, $X_I = B_I$, so the first sum is $G(B)$. If $i_k = n$, expanding along the last row, $\det X(i_1, \dots, i_{k-1}, n) = \pm\lambda \det C(i_1, \dots, i_{k-1})$, so the second sum is $\lambda^2 G(C)$. By the induction hypothesis $F(B) = G(B)$ and $F(C) = G(C)$, hence $F(X) = G(X)$. ∎

§22. The Volume of a Parametrized-Manifold

Motivation. How do we measure the area of a surface in $\mathbb{R}^3$, or the k-dimensional volume of a k-dimensional object in $\mathbb{R}^n$? Linear algebra gives the volume of a parallelepiped as $\sqrt{\det(X^T X)}$; integrating this over a parametrization defines volume, independently of the parametrization chosen.

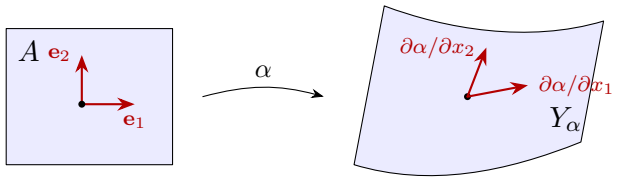


Figure. *A parametrized surface $\alpha : A \to \mathbb{R}^3$. A small square in A is mapped approximately onto the parallelogram spanned by the columns of $D\alpha$, whose area is $V(D\alpha) = \sqrt{\det(D\alpha^{\mathsf{T}} D\alpha)}$; integrating this factor over A gives the area of Y_α.*

Definition 22.1 (parametrized-manifold, volume)

Let $k \le n$, A open in $\mathbb{R}^k$, and $\alpha : A \to \mathbb{R}^n$ of class C^r ($r \ge 1$). The set $Y = \alpha(A)$ together with the map α is a **parametrized-manifold** of dimension k, denoted Y_α. Its (k-dimensional) **volume** is

$$\mathrm{v}(Y_\alpha) = \int_A V(D\alpha),$$

provided the integral exists.

Plausibility. Suppose $A = \operatorname{Int} Q$ and α extends to be C^r near Q. For a partition P of Q and a subrectangle $R = [a_1, a_1 + h_1] \times \cdots \times [a_k, a_k + h_k]$, α maps R onto a "curved rectangle" $\alpha(R) \subset Y$. The edge from $\mathbf{a}$ to $\mathbf{a} + h_i \mathbf{e}_i$ goes to a curve whose chord $\alpha(\mathbf{a} + h_i \mathbf{e}_i) - \alpha(\mathbf{a})$ is approximated to first order by $\mathbf{v}_i = D\alpha(\mathbf{a}) \cdot h_i \mathbf{e}_i = (\partial\alpha/\partial x_i) h_i$. The parallelopiped with edges $\mathbf{v}_i$ has k-volume

$$V(\mathbf{v}_1, \ldots, \mathbf{v}_k) = V(\partial\alpha/\partial x_1, \ldots, \partial\alpha/\partial x_k) \cdot (h_1 \cdots h_k) = V(D\alpha(\mathbf{a})) \cdot \mathrm{v}(R).$$

Summing over R gives a number between the lower and upper sums of $V(D\alpha)$ for P, which approximates $\int_A V(D\alpha)$ as closely as we wish.

Definition 22.2 (integral with respect to volume)

Let A be open in $\mathbb{R}^k$, $\alpha : A \to \mathbb{R}^n$ of class C^r, $Y = \alpha(A)$, and f a real-valued function continuous at each point of Y. The **integral of f over Y_α with respect to volume** is

$$\int_{Y_\alpha} f \, dV = \int_A (f \circ \alpha) V(D\alpha),$$

provided this integral exists. In particular $\mathrm{v}(Y_\alpha) = \int_{Y_\alpha} dV$.

Here dV is a meaningless symbol meaning "integral with respect to volume"; it is written dV to avoid confusion with the integral $\int_A \omega$ of a differential form (Chapter 7). The literature often uses the same symbol d for both.

Theorem 22.1

Let $g : A \to B$ be a diffeomorphism of open sets in $\mathbb{R}^k$. Let $\beta : B \to \mathbb{R}^n$ be a map of class C^r; let $Y = \beta(B)$. Let $\alpha = \beta \circ g$; then $\alpha : A \to \mathbb{R}^n$ and $Y = \alpha(A)$. If $f : Y \to \mathbb{R}$ is a continuous function, then f is integrable over Y_β if and only if it is integrable over Y_α; in this case

$$\int_{Y_\alpha} f \, dV = \int_{Y_\beta} f \, dV.$$

In particular, $\mathrm{v}(Y_\alpha) = \mathrm{v}(Y_\beta)$.

Proof. We must show $\int_B (f \circ \beta) V(D\beta) = \int_A (f \circ \alpha) V(D\alpha)$, one existing iff the other does. By the

change of variables theorem,

$$\int_B (f\circ\beta)V(D\beta) = \int_A ((f\circ\beta)\circ g)\,(V(D\beta)\circ g)\,|\det Dg|,$$

so it suffices to show $(V(D\beta)\circ g)|\det Dg| = V(D\alpha)$. With $\mathbf{y} = g(\mathbf{x})$, $D\alpha(\mathbf{x}) = D\beta(\mathbf{y})\cdot Dg(\mathbf{x})$, so

$$[V(D\alpha(\mathbf{x}))]^2 = \det(Dg(\mathbf{x})^{\mathrm{tr}}\, D\beta(\mathbf{y})^{\mathrm{tr}}\, D\beta(\mathbf{y})\, Dg(\mathbf{x})) = (\det Dg(\mathbf{x}))^2[V(D\beta(\mathbf{y}))]^2.$$

■

Example 22.1 (length of a curve)

Let A be an open interval in $\mathbb{R}^1$ and $\alpha : A \to \mathbb{R}^n$ of class C^r; Y_α is a **parametrized-curve**, and its 1-dimensional volume is its **length**:

$$\mathrm{v}(Y_\alpha) = \int_A V(D\alpha) = \int_A \left[\left(\frac{d\alpha_1}{dt}\right)^2 + \cdots + \left(\frac{d\alpha_n}{dt}\right)^2\right]^{1/2},$$

since $D\alpha$ is the column matrix of the $d\alpha_i/dt$—the familiar arc-length formula.

Example 22.2

For $\alpha(t) = (a\cos t, a\sin t)$, $0 < t < 3\pi$, the formula gives $\int_0^{3\pi}[a^2\sin^2 t + a^2\cos^2 t]^{1/2} = 3\pi a$. Since α is not one-to-one, this is not the length of the image (the circle of radius a) but the distance travelled by a particle moving by $\mathbf{x} = \alpha(t)$. Later we restrict to one-to-one parametrizations.

Example 22.3 (area of a surface)

If A is open in $\mathbb{R}^2$, Y_α is a **parametrized-surface**, and its 2-dimensional volume is its **area**. For $n = 3$, with (x, y) the general point of $\mathbb{R}^2$, $D\alpha = [\partial\alpha/\partial x \;\; \partial\alpha/\partial y]$ and (Example 21.1)

$$\mathrm{v}(Y_\alpha) = \int_A \left\|\frac{\partial\alpha}{\partial x}\times\frac{\partial\alpha}{\partial y}\right\|.$$

If $\alpha(x, y) = (x, y, f(x, y))$ with $f : A \to \mathbb{R}$ of class C^r, Y is the graph of f, and since $D\alpha$ has rows $(1, 0)$, $(0, 1)$, $(\partial f/\partial x, \partial f/\partial y)$, Theorem 21.4 gives

$$\mathrm{v}(Y_\alpha) = \int_A [1 + (\partial f/\partial x)^2 + (\partial f/\partial y)^2]^{1/2},$$

the surface-area formula of calculus.

Example 22.4 (hemisphere)

Let A be the open disc $x^2+y^2 < a^2$ and $f(x, y) = [a^2-x^2-y^2]^{1/2}$; the graph is a **hemisphere** of radius a. With $\alpha(x, y) = (x, y, f(x, y))$ one checks $V(D\alpha) = a/(a^2-x^2-y^2)^{1/2}$, so in polar coordinates

$$\mathrm{v}(Y_\alpha) = \int_B ar/(a^2 - r^2)^{1/2}, \qquad B = (0, a)\times(0, 2\pi).$$

This is improper, so Fubini does not apply directly. Integrate over $(0, a_n)\times(0, 2\pi)$ with $0 < a_n < a$ and let $a_n \to a$: $\mathrm{v}(Y_\alpha) = \lim_{n\to\infty}(-2\pi a)[(a^2 - a_n^2)^{1/2} - a] = 2\pi a^2$. (A method avoiding improper integrals is in §25.)

§23. Manifolds in $\mathbb{R}^n$

Motivation. Curves and surfaces with corners or self-intersections should be excluded, and global objects like spheres need several parametrizations. A manifold is a set covered by compatible local parametrizations, and the transition maps are diffeomorphisms by the inverse function theorem.

We restrict ourselves to submanifolds of $\mathbb{R}^n$; abstract manifolds appear in Chapter 9.

Definition 23.1 (manifold without boundary)

Let $k > 0$. Suppose M is a subspace of $\mathbb{R}^n$ such that for each $\mathbf{p} \in M$ there are a set V containing $\mathbf{p}$ and open in M, a set U open in $\mathbb{R}^k$, and a continuous map $\alpha : U \to V$ carrying U onto V in a one-to-one fashion, such that:
(1) α is of class C^r;
(2) $\alpha^{-1} : V \to U$ is continuous;
(3) $D\alpha(\mathbf{x})$ has rank k for each $\mathbf{x} \in U$. Then M is a **k-manifold without boundary** in $\mathbb{R}^n$, of class C^r, and α is a **coordinate patch** on M about $\mathbf{p}$.

Example 23.1

For $k = 1$, $D\alpha$ has rank 1 iff $D\alpha \neq \mathbf{0}$; this rules out "cusps" and "corners". If $\alpha(t) = (t^3, t^2)$, the image M has a cusp at the origin; α is C^∞ and α^{-1} is continuous, but $D\alpha(0)$ has rank 0. If $\beta(t) = (t^3, |t^3|)$, the image has a corner at the origin; β is C^2 and β^{-1} continuous, but $D\beta(0)$ has rank 0.

Example 23.2

For $k = 2$, rank 2 means the columns $\partial\alpha/\partial x_1$, $\partial\alpha/\partial x_2$ are independent at $\mathbf{a}$. Since $\partial\alpha/\partial x_j$ is the velocity of the curve $t \mapsto \alpha(\mathbf{a} + t\mathbf{e}_j)$ in M, these span a 2-dimensional "tangent plane". When the condition fails, e.g. $\alpha(x, y) = (x(x^2 + y^2),\ y(x^2 + y^2),\ x^2 + y^2)$, the image has no tangent plane at the origin (a cone-like point); α is C^∞ and α^{-1} continuous, but $D\alpha(\mathbf{0})$ has rank 0.

Example 23.3

Continuity of α^{-1} rules out other pathologies. Let $\alpha(t) = (\sin 2t)(|\cos t|, \sin t)$ for $0 < t < \pi$; its image M is a "figure eight". α is C^1 with $D\alpha$ of rank 1 and maps $(0, \pi)$ one-to-one onto M, but α^{-1} is not continuous: the image of a small interval U_0 about $\pi/2$ is not open in M (points of M near $\mathbf{0}$ need not come from points near $\pi/2$).

Example 23.4

If $\alpha : A \to \mathbb{R}^n$ is C^r on A open in $\mathbb{R}^k$ and $Y = \alpha(A)$, then Y_α is a parametrized-manifold, but Y need not be a manifold. If α is one-to-one, α^{-1} is continuous and $D\alpha$ has rank k, then Y is a manifold without boundary, covered by the single coordinate patch α.

Definition 23.2 (C^r on an arbitrary set)

Let $S \subset \mathbb{R}^k$ and $f : S \to \mathbb{R}^n$. f is **of class C^r on** S if f extends to a function $g : U \to \mathbb{R}^n$ of class C^r on an open set $U \supset S$.

Composites of C^r functions are C^r: if $f_1 : S \to \mathbb{R}^n$ and $f_2 : T \to \mathbb{R}^p$ are C^r with $f_1(S) \subset T$, with C^r extensions g_1 on U and g_2 on V, then $g_2 \circ g_1$ is a C^r extension of $f_2 \circ f_1$ on $g_1^{-1}(V) \supset S$.

Lemma 23.1

Let S be a subset of $\mathbb{R}^k$; let $f : S \to \mathbb{R}^n$. If for each $\mathbf{x} \in S$, there is a neighborhood $U_\mathbf{x}$ of $\mathbf{x}$ and a function $g_\mathbf{x} : U_\mathbf{x} \to \mathbb{R}^n$ of class C^r that agrees with f on $U_\mathbf{x} \cap S$, then f is of class C^r on S.

Proof. Cover S by the $U_\mathbf{x}$; let A be their union, and $\{\phi_i\}$ a C^∞ partition of unity on A dominated by $\{U_\mathbf{x}\}$. For each i choose $U_\mathbf{x}$ containing Support ϕ_i and let $g_i = g_\mathbf{x}$. The C^r function $\phi_i g_i : U_\mathbf{x} \to \mathbb{R}^n$ vanishes outside a closed subset of $U_\mathbf{x}$; extend it by 0 to a C^r function h_i on A. Let $g(\mathbf{x}) = \sum_i h_i(\mathbf{x})$ for $\mathbf{x} \in A$. Near each point of A, g is a finite sum of h_i's, so g is C^r on A. For $\mathbf{x} \in S$, $h_i(\mathbf{x}) = \phi_i(\mathbf{x})g_i(\mathbf{x}) = \phi_i(\mathbf{x})f(\mathbf{x})$ whenever $\phi_i(\mathbf{x}) \neq 0$, so $g(\mathbf{x}) = \sum_i \phi_i(\mathbf{x})f(\mathbf{x}) = f(\mathbf{x})$. ■

Definition 23.3 (half-space)

$\mathbb{H}^k$, **upper half-space** in $\mathbb{R}^k$, consists of the $\mathbf{x} \in \mathbb{R}^k$ with $x_k \geq 0$; $\mathbb{H}^k_+$, the **open upper half-space**, of those with $x_k > 0$.

Lemma 23.2

Let U be open in $\mathbb{H}^k$ but not in $\mathbb{R}^k$; let $\alpha : U \to \mathbb{R}^n$ be of class C^r. Let $\beta : U' \to \mathbb{R}^n$ be a C^r extension of α defined on an open set U' of $\mathbb{R}^k$. Then for $\mathbf{x} \in U$, the derivative $D\beta(\mathbf{x})$ depends only on the function α and is independent of the extension β. It follows that we may denote this derivative by $D\alpha(\mathbf{x})$ without ambiguity.

Proof. $\partial\beta_i/\partial x_j$ at $\mathbf{x}$ is the limit of $[\beta(\mathbf{x} + h\mathbf{e}_j) - \beta(\mathbf{x})]/h$ as $h \to 0$; it may be computed letting $h \to 0$ through positive values. If $\mathbf{x} \in \mathbb{H}^k$, so is $\mathbf{x} + h\mathbf{e}_j$ for $h > 0$, and there β agrees with α; so $D\beta(\mathbf{x})$ depends only on α. ■

Definition 23.4 (k-manifold)

Let $k > 0$. A **k-manifold in $\mathbb{R}^n$ of class C^r** is a subspace M of $\mathbb{R}^n$ such that for each $\mathbf{p} \in M$ there are an open set V of M containing $\mathbf{p}$, a set U open in either $\mathbb{R}^k$ or $\mathbb{H}^k$, and a continuous map $\alpha : U \to V$ carrying U onto V in a one-to-one fashion, such that:
(1) α is of class C^r;
(2) $\alpha^{-1} : V \to U$ is continuous;
(3) $D\alpha(\mathbf{x})$ has rank k for each $\mathbf{x} \in U$. The map α is a **coordinate patch** on M about $\mathbf{p}$. A discrete collection of points in $\mathbb{R}^n$ is a **0-manifold**.

A manifold without boundary is the special case where all coordinate patches have domains open in $\mathbb{R}^k$. A k-manifold has two kinds of points: those with neighborhoods like open k-balls, and those with neighborhoods only like open half-balls; the latter form the *boundary* (§24).

Lemma 23.3

Let M be a manifold in $\mathbb{R}^n$, and let $\alpha : U \to V$ be a coordinate patch on M. If U_0 is a subset of U that is open in U, then the restriction of α to U_0 is also a coordinate patch on M.

Proof. Since U_0 is open in U and α^{-1} is continuous, $V_0 = \alpha(U_0)$ is open in V, hence in M; and U_0 is open in $\mathbb{R}^k$ or $\mathbb{H}^k$ according as U is. $\alpha|U_0$ maps U_0 one-to-one onto V_0, is C^r, its inverse is a restriction of α^{-1}, hence continuous, and its derivative has rank k. ■

This fails without continuity of α^{-1}: in Example 23.3, $\alpha|U_0$ is not a coordinate patch since its image is not open in M.

§24. The Boundary of a Manifold

Motivation. Stokes' theorem relates an integral over a manifold to one over its boundary, so we need the boundary itself to be a manifold of one lower dimension. Using half-spaces as local models makes this automatic.

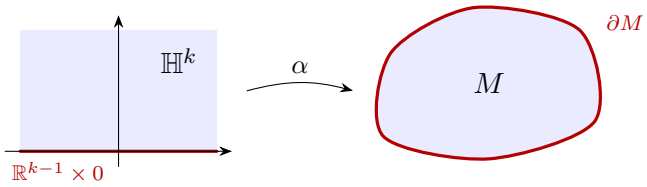


Figure. *Points of a k-manifold M with boundary have neighbourhoods modelled either on $\mathbb{R}^k$ (interior points) or on the half-space $\mathbb{H}^k$ (boundary points); the boundary ∂M is a $(k-1)$-manifold without boundary.*

Theorem 24.1

Let M be a k-manifold in $\mathbb{R}^n$, of class C^r. Let $\alpha_0 : U_0 \to V_0$ and $\alpha_1 : U_1 \to V_1$ be coordinate patches on M, with $W = V_0 \cap V_1$ non-empty. Let $W_i = \alpha_i^{-1}(W)$. Then the map

$$\alpha_1^{-1} \circ \alpha_0 : W_0 \to W_1$$

is of class C^r, and its derivative is non-singular.

We call $\alpha_1^{-1} \circ \alpha_0$ the **transition function** between α_0 and α_1.

Proof. It suffices to show: if $\alpha : U \to V$ is a coordinate patch on M, then $\alpha^{-1} : V \to \mathbb{R}^k$ is of class C^r as a map of the subset V of $\mathbb{R}^n$. Then $\alpha_1^{-1} \circ \alpha_0$ and $\alpha_0^{-1} \circ \alpha_1$ are C^r, inverse to each other, and the chain rule shows their derivatives are non-singular.

By Lemma 23.1 it suffices to show α^{-1} is locally C^r. Let $\mathbf{p}_0 \in V$, $\mathbf{x}_0 = \alpha^{-1}(\mathbf{p}_0)$. Suppose U is open in $\mathbb{H}^k$ but not $\mathbb{R}^k$; extend α to a C^r map β on an open set U' of $\mathbb{R}^k$. $D\alpha(\mathbf{x}_0)$ has rank k, so some k rows are independent; say the first k. Let $\pi : \mathbb{R}^n \to \mathbb{R}^k$ project onto the first k coordinates. Then $g = \pi \circ \beta$ maps U' into $\mathbb{R}^k$ with $Dg(\mathbf{x}_0)$ non-singular, so by the inverse function theorem g is a C^r diffeomorphism of an open set W of $\mathbb{R}^k$ about $\mathbf{x}_0$ with an open set of $\mathbb{R}^k$. We claim $h = g^{-1} \circ \pi$, which is C^r, extends α^{-1} near $\mathbf{p}_0$. $U_0 = W \cap U$ is open in U, so $V_0 = \alpha(U_0)$ is open in V, hence there is an open $A \subset \mathbb{R}^n$ with $A \cap V = V_0$; shrink A into the domain of h (intersect with $\pi^{-1}(g(W))$). For $\mathbf{p} \in A \cap V = V_0$, with $\mathbf{x} = \alpha^{-1}(\mathbf{p})$,

$$h(\mathbf{p}) = g^{-1}(\pi(\alpha(\mathbf{x}))) = g^{-1}(g(\mathbf{x})) = \mathbf{x} = \alpha^{-1}(\mathbf{p}).$$

If U is open in $\mathbb{R}^k$, take $U' = U$, $\beta = \alpha$; the argument is unchanged. ∎

Definition 24.1 (interior and boundary points)

Let M be a k-manifold in $\mathbb{R}^n$ and $\mathbf{p} \in M$. If there is a coordinate patch $\alpha : U \to V$ on M about $\mathbf{p}$ with U open in $\mathbb{R}^k$, $\mathbf{p}$ is an **interior point** of M; otherwise $\mathbf{p}$ is a **boundary point**. The set of boundary points is the **boundary** ∂M of M.

These terms are unrelated to the topological notions $\operatorname{Int} S$, $\operatorname{Bd} S$, $\operatorname{Ext} S$ of a subset $S \subset \mathbb{R}^n$. For a manifold we write ∂M for its boundary and $M - \partial M$ for its interior.

Lemma 24.2

Let M be a k-manifold in $\mathbb{R}^n$; let $\alpha : U \to V$ be a coordinate patch about the point $\mathbf{p}$ of M.

(a) If U is open in $\mathbb{R}^k$, then $\mathbf{p}$ is an interior point of M.

(b) If U is open in $\mathbb{H}^k$ and if $\mathbf{p} = \alpha(\mathbf{x}_0)$ for $\mathbf{x}_0 \in \mathbb{H}^k_+$, then $\mathbf{p}$ is an interior point of M.

(c) If U is open in $\mathbb{H}^k$ and $\mathbf{p} = \alpha(\mathbf{x}_0)$ for $\mathbf{x}_0 \in \mathbb{R}^{k-1} \times 0$, then $\mathbf{p}$ is a boundary point of M.

Proof. (a) is the definition.

(b) Let $U_0 = U \cap \mathbb{H}^k_+$, open in $\mathbb{R}^k$, and $V_0 = \alpha(U_0)$; then $\alpha|U_0$ is a coordinate patch about $\mathbf{p}$ (Lemma 23.3) with domain open in $\mathbb{R}^k$.

(c) Let $\alpha_0 : U_0 \to V_0$ be a patch about $\mathbf{p}$ with U_0 open in $\mathbb{H}^k$ and $\mathbf{p} = \alpha_0(\mathbf{x}_0)$, $\mathbf{x}_0 \in \mathbb{R}^{k-1} \times 0$. Suppose there were a patch $\alpha_1 : U_1 \to V_1$ about $\mathbf{p}$ with U_1 open in $\mathbb{R}^k$. $W = V_0 \cap V_1$ is open in M; $W_0 = \alpha_0^{-1}(W)$ is open in $\mathbb{H}^k$ and contains $\mathbf{x}_0$, and $W_1 = \alpha_1^{-1}(W)$ is open in $\mathbb{R}^k$. By Theorem 24.1, $\alpha_0^{-1} \circ \alpha_1 : W_1 \to W_0$ is C^r, one-to-one onto W_0, with non-singular derivative. By Theorem 8.2 its image W_0 is open in $\mathbb{R}^k$. But $W_0 \subset \mathbb{H}^k$ contains the point $\mathbf{x}_0$ of $\mathbb{R}^{k-1} \times 0$, so it is not open in $\mathbb{R}^k$—a contradiction. ■

$\mathbb{H}^k$ is itself a k-manifold in $\mathbb{R}^k$, and by this lemma $\partial\mathbb{H}^k = \mathbb{R}^{k-1} \times 0$.

Theorem 24.3

Let M be a k-manifold in $\mathbb{R}^n$, of class C^r. If ∂M is non-empty, then ∂M is a $k-1$ manifold without boundary in $\mathbb{R}^n$ of class C^r.

Proof. Let $\mathbf{p} \in \partial M$ and $\alpha : U \to V$ a patch about $\mathbf{p}$. Then U is open in $\mathbb{H}^k$ and $\mathbf{p} = \alpha(\mathbf{x}_0)$ with $\mathbf{x}_0 \in \partial\mathbb{H}^k$. By Lemma 24.2, α maps points of $U \cap \mathbb{H}^k_+$ to interior points of M and points of $U \cap \partial\mathbb{H}^k$ to points of ∂M. So α restricted to $U \cap \partial\mathbb{H}^k$ maps it one-to-one onto the open set $V_0 = V \cap \partial M$ of ∂M. Let U_0 be the open set of $\mathbb{R}^{k-1}$ with $U_0 \times 0 = U \cap \partial\mathbb{H}^k$, and define $\alpha_0(\mathbf{x}) = \alpha(\mathbf{x}, 0)$ for $\mathbf{x} \in U_0$. Then $\alpha_0 : U_0 \to V_0$ is a coordinate patch on ∂M: it is C^r because α is; $D\alpha_0(\mathbf{x})$ consists of the first $k-1$ columns of $D\alpha(\mathbf{x}, 0)$, so it has rank $k-1$; and α_0^{-1} is the restriction to V_0 of α^{-1} followed by projection of $\mathbb{R}^k$ onto its first $k-1$ coordinates, hence continuous. ■

The patch α_0 on ∂M is said to be obtained by **restricting** the patch α on M.

Theorem 24.4

Let $\mathcal{O}$ be open in $\mathbb{R}^n$; let $f : \mathcal{O} \to \mathbb{R}$ be of class C^r. Let M be the set of points $\mathbf{x}$ for which $f(\mathbf{x}) = 0$; let N be the set of points for which $f(\mathbf{x}) \geq 0$. Suppose M is non-empty and $Df(\mathbf{x})$ has rank 1 at each point of M. Then N is an n-manifold in $\mathbb{R}^n$ and $\partial N = M$.

Proof. If $\mathbf{p} \in N$ with $f(\mathbf{p}) > 0$, let $U = \{\mathbf{x} : f(\mathbf{x}) > 0\}$, open in $\mathbb{R}^n$; the identity $\alpha : U \to U$ is a coordinate patch on N about $\mathbf{p}$ with domain open in $\mathbb{R}^n$. If $f(\mathbf{p}) = 0$, then $Df(\mathbf{p}) \neq 0$, so some $D_i f(\mathbf{p}) \neq 0$; say $D_n f(\mathbf{p}) \neq 0$. Define $F : \mathcal{O} \to \mathbb{R}^n$ by $F(\mathbf{x}) = (x_1, \ldots, x_{n-1}, f(\mathbf{x}))$. Then

$$DF = \begin{bmatrix} I_{n-1} & 0 \\ * & D_n f \end{bmatrix}$$

is non-singular at $\mathbf{p}$, so F is a diffeomorphism of a neighborhood A of $\mathbf{p}$ with an open set B of $\mathbb{R}^n$. F carries $A \cap N$ onto $B \cap \mathbb{H}^n$ (as $\mathbf{x} \in N$ iff $f(\mathbf{x}) \geq 0$) and $A \cap M$ onto $B \cap \partial\mathbb{H}^n$ (as $\mathbf{x} \in M$ iff $f(\mathbf{x}) = 0$). So $F^{-1} : B \cap \mathbb{H}^n \to A \cap N$ is a coordinate patch on N about $\mathbf{p}$ with domain open in $\mathbb{H}^n$, and $\mathbf{p} = F^{-1}(F(\mathbf{p}))$ with $F(\mathbf{p}) \in \partial\mathbb{H}^n$. By Lemma 24.2, $\partial N = M$. ■

Definition 24.2 (ball and sphere)

$B^n(a) = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x}\| \le a\}$ is the n-**ball** and $S^{n-1}(a) = \{\mathbf{x} : \|\mathbf{x}\| = a\}$ the $(n-1)$-**sphere** of radius a.

Corollary 24.5

The n-ball $B^n(a)$ is an n-manifold in $\mathbb{R}^n$ of class C^∞, and $S^{n-1}(a) = \partial B^n(a)$.

Proof. Apply Theorem 24.4 to $f(\mathbf{x}) = a^2 - \|\mathbf{x}\|^2$; $Df(\mathbf{x}) = [-2x_1 \ \cdots \ -2x_n]$ is nonzero at each point of $S^{n-1}(a)$. ■

§25. Integrating a Scalar Function over a Manifold

Motivation. To integrate over a manifold covered by several patches we cut the function into pieces supported in single patches, using partitions of unity, and add the results. The result does not depend on the choices made.

For simplicity we restrict to compact manifolds M; the general case can be handled as in §16.

Definition 25.1 (integral, support in one patch)

Let M be a compact k-manifold in $\mathbb{R}^n$ of class C^r, and $f : M \to \mathbb{R}$ continuous with $C = \operatorname{Support} f$ (compact). Suppose there is a coordinate patch $\alpha : U \to V$ on M with $C \subset V$; since $\alpha^{-1}(C)$ is compact we may assume U bounded. Define

$$\int_M f\,dV = \int_{\operatorname{Int} U} (f \circ \alpha) V(D\alpha),$$

where $\operatorname{Int} U = U$ if U is open in $\mathbb{R}^k$, and $\operatorname{Int} U = U \cap \mathbb{H}^k_+$ if U is open in $\mathbb{H}^k$ but not in $\mathbb{R}^k$.

This exists as an ordinary integral (hence as an extended one): $F = (f \circ \alpha)V(D\alpha)$ is continuous on U and vanishes outside the compact set $\alpha^{-1}(C)$, so it is bounded. If U is open in $\mathbb{R}^k$, F vanishes near each point of $\operatorname{Bd} U$; otherwise F vanishes near each point of $\operatorname{Bd} U$ not in $\partial\mathbb{H}^k$, a set of measure zero. Either way F is integrable over U and over $\operatorname{Int} U$.

Lemma 25.1

If the support of f can be covered by a single coordinate patch, the integral $\int_M f\,dV$ is well-defined, independent of the choice of coordinate patch.

Proof. First, if $\alpha : U \to V$ contains $\operatorname{Support} f$ and $W \subset U$ is open with $\alpha(W) \supset \operatorname{Support} f$, then $\int_{\operatorname{Int} W}(f \circ \alpha)V(D\alpha) = \int_{\operatorname{Int} U}(f \circ \alpha)V(D\alpha)$: the integrand vanishes outside W, then apply Theorem 13.6. Now let $\alpha_i : U_i \to V_i$ $(i = 0, 1)$ be patches with $V_0, V_1 \supset \operatorname{Support} f$. Let $W = V_0 \cap V_1$, $W_i = \alpha_i^{-1}(W)$. By the preceding remark it suffices to show $\int_{\operatorname{Int} W_0}(f \circ \alpha_0)V(D\alpha_0) = \int_{\operatorname{Int} W_1}(f \circ \alpha_1)V(D\alpha_1)$. Since $\alpha_1^{-1} \circ \alpha_0 : \operatorname{Int} W_0 \to \operatorname{Int} W_1$ is a diffeomorphism (Theorem 24.1, and Lemma 24.2 for interiors), this follows from Theorem 22.1. ■

Lemma 25.2

Let M be a compact k-manifold in $\mathbb{R}^n$, of class C^r. Given a covering of M by coordinate patches, there exists a finite collection of C^∞ functions $\phi_1, \dots, \phi_l$ mapping $\mathbb{R}^n$ into $\mathbb{R}$ such that:

(1) $\phi_i(\mathbf{x}) \geq 0$ for all $\mathbf{x}$.
(2) Given i, the support of ϕ_i is compact and there is a coordinate patch $\alpha_i : U_i \to V_i$ belonging to the given covering such that $(\text{Support}\,\phi_i) \cap M \subset V_i$.
(3) $\sum \phi_i(\mathbf{x}) = 1$ for $\mathbf{x} \in M$.

We call $\{\phi_1, \ldots, \phi_l\}$ a **partition of unity on** M dominated by the given collection of coordinate patches.

Proof. For each patch $\alpha : U \to V$ in the covering choose an open set A_V of $\mathbb{R}^n$ with $A_V \cap M = V$. Let A be the union of the A_V, and choose a C^∞ partition of unity on A with compact supports dominated by $\{A_V\}$ (Theorem 16.3). By local finiteness and compactness of M, all but finitely many of these functions vanish identically on M; let $\phi_1, \ldots, \phi_l$ be those that do not. ■

Definition 25.2 (integral over a compact manifold; volume)

Let M be a compact k-manifold in $\mathbb{R}^n$ of class C^r and $f : M \to \mathbb{R}$ continuous. Choose a partition of unity $\phi_1, \ldots, \phi_l$ on M dominated by the collection of all coordinate patches on M. Define

$$\int_M f\,dV = \sum_{i=1}^{l} \Big[\int_M (\phi_i f)\,dV \Big],$$

and the (k-dimensional) **volume** of M by $\mathrm{v}(M) = \int_M 1\,dV$.

If Support f lies in a single patch $\alpha : U \to V$, this agrees with Definition 25.1: with $A = \operatorname{Int} U$,

$$\sum_{i=1}^{l} \Big[\int_M (\phi_i f)\,dV \Big] = \sum_{i=1}^{l} \Big[\int_A (\phi_i \circ \alpha)(f \circ \alpha) V(D\alpha) \Big] = \int_A \Big[\sum_i (\phi_i \circ \alpha) \Big] (f \circ \alpha) V(D\alpha) = \int_A (f \circ \alpha) V(D\alpha),$$

since $\sum_i (\phi_i \circ \alpha) = 1$ on A. It is independent of the partition of unity: if $\psi_1, \ldots, \psi_m$ is another, then since Support $\psi_j f$ lies in a single patch, the computation just given (with $\psi_j f$ in place of f) gives $\sum_i \int_M \phi_i \psi_j f\,dV = \int_M \psi_j f\,dV$. Summing over j, $\sum_j \sum_i \int_M \phi_i \psi_j f\,dV = \sum_j \int_M \psi_j f\,dV$, and by symmetry the double sum also equals $\sum_i \int_M \phi_i f\,dV$.

Theorem 25.3

Let M be a compact k-manifold in $\mathbb{R}^n$, of class C^r. Let $f, g : M \to \mathbb{R}$ be continuous. Then

$$\int_M (af + bg)\,dV = a \int_M f\,dV + b \int_M g\,dV.$$

Proof [Supplement]. For f, g supported in one patch this is linearity of the integral over $\operatorname{Int} U$; in general apply it to each $\phi_i f, \phi_i g$ and sum. ■

This definition is good for theory but not for computation: to integrate over S^{n-1}, one breaks it into pieces, integrates over each, and adds. The next theorem justifies this.

Definition 25.3 (measure zero in M)

Let M be a compact k-manifold in $\mathbb{R}^n$ of class C^r. A subset D of M has **measure zero in** M if it can be covered by countably many coordinate patches $\alpha_i : U_i \to V_i$ such that each $D_i = \alpha_i^{-1}(D \cap V_i)$ has measure zero in $\mathbb{R}^k$.

Equivalently, $\alpha^{-1}(D \cap V)$ has measure zero in $\mathbb{R}^k$ for *every* patch $\alpha : U \to V$: it is the union of the sets $\alpha^{-1}(D \cap V \cap V_i)$, and each is the image of a subset of D_i under the C^r map $\alpha^{-1} \circ \alpha_i$, hence has measure zero (Lemma 18.1).

Theorem 25.4

Let M be a compact k-manifold in $\mathbb{R}^n$, of class C^r. Let $f : M \to \mathbb{R}$ be a continuous function. Suppose that $\alpha_i : A_i \to M_i$, for $i = 1, \dots, N$, is a coordinate patch on M, such that A_i is open in $\mathbb{R}^k$ and M is the disjoint union of the open sets $M_1, \dots, M_N$ of M and a set K of measure zero in M. Then

$$\int_M f\,dV = \sum_{i=1}^{N} \Big[\int_{A_i} (f \circ \alpha_i) V(D\alpha_i) \Big]. \tag{$*$}$$

So $\int_M f\,dV$ can be computed by breaking M into pieces that are parametrized-manifolds and integrating over each.

Proof. Both sides of $(*)$ are linear in f, so we may assume $C = \operatorname{Support} f$ lies in a single patch $\alpha : U \to V$, U bounded. Then $\int_M f\,dV = \int_{\operatorname{Int} U} (f \circ \alpha) V(D\alpha)$.

Step 1. Let $W_i = \alpha^{-1}(M_i \cap V)$ and $L = \alpha^{-1}(K \cap V)$. Then W_i is open in $\mathbb{R}^k$, L has measure zero in $\mathbb{R}^k$, and U is the disjoint union of L and the W_i. We show $\int_M f\,dV = \sum_i \int_{W_i} F$, where $F = (f \circ \alpha) V(D\alpha)$. These are ordinary integrals: F is bounded and vanishes near each point of $\operatorname{Bd} W_i$ not in L. Then

$$\sum_i \Big[\int_{W_i} F \Big] = \int_{(\operatorname{Int} U) - L} F = \int_{\operatorname{Int} U} F = \int_M f\,dV,$$

by additivity, since L has measure zero, and by definition.

Step 2. We show $\int_{W_i} F = \int_{A_i} F_i$, with $F_i = (f \circ \alpha_i) V(D\alpha_i)$. The map $\alpha_i^{-1} \circ \alpha$ is a diffeomorphism of W_i onto the open set $B_i = \alpha_i^{-1}(M_i \cap V)$ of $\mathbb{R}^k$, so by the change of variables theorem, as in Theorem 22.1, $\int_{W_i} F = \int_{B_i} F_i$. These may not be ordinary integrals, so care is needed. Since C is closed in M, $\alpha_i^{-1}(C)$ is closed in A_i, so $D_i = A_i - \alpha_i^{-1}(C)$ is open in $\mathbb{R}^k$, and F_i vanishes on D_i. Additivity of the extended integral gives

$$\int_{A_i} F_i = \int_{B_i} F_i + \int_{D_i} F_i - \int_{B_i \cap D_i} F_i;$$

the last two integrals vanish ($A_i = B_i \cup D_i$ since $\alpha_i^{-1}(C) \subset B_i$). ∎

Example 25.1

The open upper hemisphere of $S^2(a)$ has area $2\pi a^2$ (Example 22.4). The reflection $(x, y, z) \mapsto (x, y, -z)$ is an isometry of $\mathbb{R}^3$, so the open lower hemisphere also has area $2\pi a^2$. The two hemispheres make up all of the sphere except a set of measure zero (the equator), so by Theorem 25.4, $S^2(a)$ has area $4\pi a^2$.

Example 25.2

A method with no improper integrals. For $|z_0| < a$, the plane $z = z_0$ meets $S^2(a)$ in the circle $x^2 + y^2 = a^2 - z_0^2$. This suggests parametrizing $S^2(a)$ by $\alpha : A \to \mathbb{R}^3$,

$$\alpha(t, z) = ((a^2 - z^2)^{1/2} \cos t,\ (a^2 - z^2)^{1/2} \sin t,\ z), \qquad A = \{(t, z) : 0 < t < 2\pi,\ |z| < a\}.$$

α is a coordinate patch covering all of $S^2(a)$ except a great-circle arc, which has measure zero. Then

$$D\alpha = \begin{bmatrix} -(a^2 - z^2)^{1/2} \sin t & (-z \cos t)/(a^2 - z^2)^{1/2} \\ (a^2 - z^2)^{1/2} \cos t & (-z \sin t)/(a^2 - z^2)^{1/2} \\ 0 & 1 \end{bmatrix},$$

and one checks $V(D\alpha) = a$. Hence $\mathrm{v}(S^2(a)) = \int_A a = 4\pi a^2$.